

HW 9.

Exercise 1. Fix some α with $0 < \alpha < \pi/2$ and some $h > 0$. Let

$$X(\theta, t) = (t \sin \alpha \cos \theta, t \sin \alpha \sin \theta, t \cos \alpha),$$

where $\theta \in [0, 2\pi]$ and $t \in [0, h \sec \alpha]$. This parametrizes a cone of height h . Compute the tangent vectors $\partial X/\partial \theta$ and $\partial X/\partial t$ as well as their cross product. Find the equation for the surface in rectangular/Cartesian coordinates.

Exercise 2. Fix $h, a > 0$ Let

$$X(t, \theta) = (at \cos \theta, at \sin \theta, t^2)$$

where $\theta \in [0, 2\pi]$ and $t \in [0, h]$. This parametrizes a paraboloid. Compute the tangent vectors $\partial X/\partial \theta$ and $\partial X/\partial t$ as well as their cross product. Find the equation for the surface in rectangular/Cartesian coordinates. What surface do we obtain if we replace t^2 with t in the z coordinate?

Exercise 3. Compute the surface area of the torus parametrized by

$$X(\theta, \varphi) = ((a + b \cos \varphi) \cos \theta, (a + b \cos \varphi) \sin \theta, b \sin \varphi)$$

where $0 \leq \varphi < 2\pi$ and $0 \leq \theta < 2\pi$. (Hint: the cross product computation is a little messy, but using the identity $\sin^2 x + \cos^2 x = 1$ a couple of times, you can simplify the expression for $\left\| \frac{\partial X}{\partial \varphi} \times \frac{\partial X}{\partial \theta} \right\|$ considerably).

Exercise 4. Let $\varphi : \mathbb{R}^3 \rightarrow \mathbb{R}$ be a smooth function. Show that

$$\text{curl}(\text{grad } \varphi) = 0.$$

Exercise 5. Let $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a smooth vector field. Show that $\text{div } \text{curl } F = 0$.

Exercise 6. Compute the integral

$$\iint_S \text{curl } F \cdot \vec{n} d\sigma$$

where $F = (-y, x^2, z^3)$, and S is the surface

$$x^2 + y^2 + z^2 = 1, \quad -\frac{1}{2} \leq z \leq 1.$$

[Hint: by Stokes' theorem, the integral above of any two surfaces that share the same boundary will have the same value, so one can deform this surface (while holding the boundary fixed) to a much simpler one and compute the integral of $\text{curl } F$ on the deformed surface instead.]

Exercise 7. Let C be a closed curve which is the boundary of a surface S . Let f and g be scalar-valued functions. Show that

$$\int_C (f \text{ grad } g) \cdot dC = \iint_S [(\text{grad } f) \times (\text{grad } g)] \cdot \mathbf{n} d\sigma.$$

Show consequently that

$$\int_C (f \text{ grad } g + g \text{ grad } f) \cdot dC = 0.$$